

GHD Minimum Note: review of §1–§2.5

De Nardis, Bernard, Doyon, *Diffusion in generalized hydrodynamics and quasiparticle scattering*, SciPost Phys. **6**, 049 (2019), arXiv:1812.00767v4.

Purpose. Recover the physical story, essential equations, and important qualifications from our discussions. This is a selective review, with no page limit. Detailed derivations are in the detailed study notes; source audits are in the companion review.

1. The story: what problem are we solving?

Given a slowly varying initial state of an interacting integrable system, predict how its conserved-density profiles **move and broaden**. Euler GHD already predicts propagation through effective quasiparticle velocities. The missing ingredient is the leading diffusive correction and its microscopic origin:

$$\partial_t \bar{q}_i + \partial_x F_i(\bar{q}) = \frac{1}{2} \partial_x \left[\mathfrak{D}_i^j(\bar{q}) \partial_x \bar{q}_j \right]. \quad (2.10)$$

The target is the state-dependent matrix $\mathfrak{D}$. Writing this equation identifies it; calculating it makes the equation predictive. Effective velocities already include interactions through dressing. Diffusion describes additional broadening around propagation; it does not require quasiparticles to decay.

For an isolated linear mode, the illustrative dispersion $\omega_a(k) = v_a k - i\gamma_a k^2/2 + O(k^3)$ means a centre moving as $v_a t$ and variance growth $\gamma_a t$. A density can excite many modes: its total width can include ballistic separation, not just $\sqrt{t}$ broadening.

The route through the paper

Part Job in the argument

§2	Build the bridge from microscopic equilibrium correlations to macroscopic diffusion.
§3	Describe stationary states and correlations using quasiparticles and thermodynamic form factors.
§4	Calculate diffusion from two-particle-hole contributions, interpreted through scattering.
§5	Insert diffusion into GHD; study profile evolution and entropy production.
§6	Apply the theory to spin diffusion in gapped XXZ.

Only §1–§2.5 have been worked through. Later sections are a roadmap. Their results rely on hydrodynamic and form-factor assumptions, not a derivation of hydrodynamic closure from first principles.

What completing Section 2 accomplished

$$\boxed{\text{Find } \mathfrak{D} \quad \longleftarrow \quad \mathfrak{D} = \mathfrak{L} C^{-1} \quad \longleftarrow \quad \text{calculate current correlations and } C.}$$

This identification uses the paper’s PT gauge and an invertible susceptibility sector. We have established the bridge within its stated framework; we have not yet calculated an interacting model’s diffusion matrix.

2. Define the missing coefficient [§2.1]

Exact input: conserved charges and continuity,

$$Q_i = \int dx q_i(x, t), \quad \partial_t q_i + \partial_x j_i = 0, \quad \partial_t \bar{q}_i + \partial_x \bar{j}_i = 0.$$

The last equation does not determine the current. **Hydrodynamic assumption:** after local relaxation, slow density profiles determine local means. Locality and slow variation permit the current expansion

$$\boxed{\bar{j}_i = F_i(\bar{q}) - \frac{1}{2} \mathfrak{D}_i^j(\bar{q}) \partial_x \bar{q}_j + O(\partial_x^2).} \quad (2.9)$$

This is a closure for expectations, not an operator identity. Its first term is Euler transport; its first-gradient term supplies the second-gradient term in the density equation.

Where does the equation of state F come from?

Use the *family* of homogeneous stationary GGEs, Eq. (2.3):

$$\rho_\beta = Z^{-1} e^{-\sum_i \beta_i Q_i}, \quad \bar{q}_i(\beta) = \langle q_i \rangle_{\rho_\beta}, \quad \bar{j}_i(\beta) = \langle j_i \rangle_{\rho_\beta}.$$

Where densities locally parametrize this family, eliminate β to define

$$F_i(\bar{q}) = \bar{j}_i(\beta(\bar{q})), \quad \boxed{A_i^j = \frac{\partial F_i}{\partial \bar{q}_j}}. \quad (2.20)$$

One GGE gives one value of F ; nearby homogeneous GGEs determine its derivative A . Thus A can be nonzero even when the chosen state's spatial gradients vanish. Conservation alone does not determine F .

A fixed inhomogeneous state and a family of local homogeneous states are different: differentiating $\langle o(x) \rangle_\rho$ at fixed ρ acts on o ; differentiating $\langle o(x) \rangle_{\rho_x}$ also changes the state.

Recall: Why does a spatially uniform state still have a nonzero flux Jacobian?

3. Identify transport from fluctuations [§2.2]

In a homogeneous stationary state, define ordered connected correlations:

$$S_{ij}(x, t) = \langle q_i(x, t) q_j(0, 0) \rangle^c, \quad (2.11)$$

$$K_{ij}(t) = \int dx \langle j_i(x, t) j_j(0, 0) \rangle^c,$$

$$M_n(t) = \int dx x^n S(x, t), \quad C = M_0(t). \quad (2.12)$$

Connected means $\langle AB \rangle^c = \langle AB \rangle - \langle A \rangle \langle B \rangle$. Conservation fixes the total correlation weight C , not its shape. C measures the strength of equilibrium density fluctuations.

Why currents tell us about density spreading

Apply continuity to both density insertions and use translation invariance. With G the local current-current correlation,

$$\partial_t^2 S = \partial_x^2 G \implies \boxed{M_2''(t) = 2K(t)}$$

by two spatial integrations by parts and vanishing boundary terms. Integrating this relation produces the exact sum rule (2.13). It is the bridge from current fluctuations to density spreading.

Separate persistent transport from the remainder

Componentwise, Eqs. (2.15)–(2.16) define

$$\boxed{D = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T du K(u), \quad \mathfrak{L} = \lim_{T \rightarrow \infty} \int_{-T}^T du [K(u) - D].}$$

Under the paper's convergence assumptions,

$$\boxed{\frac{M_2(t) + M_2(-t)}{2} - M_2(0) = Dt^2 + \mathfrak{L}t + o(t), \quad t \rightarrow +\infty.} \quad (2.14)$$

M_2 is an uncentered correlation moment; it is not generally a probability variance.

Symbol	Meaning	Where it enters
D	Drude matrix: persistent current correlations	Ballistic t^2 growth
$\mathfrak{L}$	Onsager matrix: integrated residual current correlations	Subleading t growth
$\mathfrak{D}$	Diffusion matrix: constitutive gradient coefficient	Mean-current closure (2.9)

Recall: If $K(t) = k \neq 0$ at every time, then $D = k$ and $\mathfrak{L} = 0$: transport can be entirely persistent.

4. Match correlations to the hydrodynamic coefficient [§2.3]

The response bridge [Appendix B]

Perturb the *initial ensemble* with a positive response source η . In the commuting case, differentiating the normalized ensemble gives

$$\frac{\delta \langle o(x, t) \rangle}{\delta \eta_j(y)} = \langle o(x, t) q_j(y, 0) \rangle^c.$$

For noncommuting quantum sources, an exponential-source derivative generally gives a Kubo–Mori insertion. Appendix B uses an ordered-response identity as an input at this order; it must not be presented as an automatic identity for arbitrary quantum sources.

Differentiate the closure around a homogeneous background:

$$\frac{\delta \bar{j}_i(x, t)}{\delta \bar{q}_k(y, t)} = A_i^k \delta(x - y) - \frac{1}{2} \mathfrak{D}_i^k \partial_x \delta(x - y).$$

A variation of $\mathfrak{D}$ multiplies the zero background gradient, so contributes nothing to this linearization. The response chain rule and exact continuity then give

$$\boxed{\partial_t S = -A \partial_x S + \frac{1}{2} \mathfrak{D} \partial_x^2 S, \quad t > 0.} \quad (2.19)$$

This is a homogeneous-background correlation equation. It is not the full nonlinear equation for an arbitrary profile.

Negative time: keep the order and follow the later insertion

A response imposed at time 0 does not reconstruct an earlier state. Define

$$J_{ij} = \langle j_i(x, t) q_j(0, 0) \rangle^c, \quad H_{ij} = \langle q_i(x, t) j_j(0, 0) \rangle^c.$$

Continuity at the two insertions gives $\partial_x J = \partial_x H = -\partial_t S$; clustering implies $J = H$, Eq. (B.5). Translation preserves operator order:

$$H_{ij}(x, t) = \langle q_i(0, 0) j_j(-x, -t) \rangle^c.$$

For $t < 0$, $-t > 0$: the current is now the later insertion and carries the *second* charge index. The corresponding ordered response gives

$$J_{ij} = A_j^k S_{ik} + \frac{1}{2} \mathfrak{D}_j^k \partial_x S_{ik},$$

where the gradient sign comes from $\partial_{(-x)} = -\partial_x$. Hence

$$\boxed{\partial_t S = -(\partial_x S) A^\top - \frac{1}{2} (\partial_x^2 S) \mathfrak{D}^\top, \quad t < 0.} \quad (2.19)$$

The transpose records which index evolves. It does not come from exchanging the operators. The negative-time branch is not a forward diffusion initial-value problem.

Moments extract exactly the coefficients we need

Let $E = M_1(0) = \int dx x S(x, 0)$, Eq. (2.22). Then

$$M_0 = C, \quad M_1(t) = ACt + E, \quad AC = CA^\top,$$

using Eqs. (2.21), (2.23). For $t > 0$, integrating Eqs. (2.24)–(2.25) yields, at the retained hydrodynamic orders,

$$\begin{aligned} N(t) &:= M_2(t) - M_2(0) = A^2 C t^2 + (2AE + \mathfrak{D}C)t, \\ N(-t) &= A^2 C t^2 + (C\mathfrak{D}^\top - 2EA^\top)t. \end{aligned}$$

Matching the symmetrized moment to (2.14) gives

$$\boxed{D = A^2 C = ACA^\top, \quad \mathfrak{L} = \frac{1}{2}(\mathfrak{D}C + C\mathfrak{D}^\top) + AE - EA^\top.}$$

The displayed E coefficient follows direct integration; it differs by an *apparent* factor of two from printed (2.26)–(2.27). Audit details belong in the companion review. This discrepancy disappears when $E = 0$.

Our progress: correlation-defined transport is related to the constitutive coefficient, but E and a local-density ambiguity obstruct the simple formula $\mathfrak{L} = \mathfrak{D}C$.

5. Fix the local-density convention [§2.4]

What “gauge” means here

Different local densities can represent the same integrated charge. Equation (2.28) permits

$$\boxed{q'_i = q_i + \partial_x o_i, \quad j'_i = j_i - \partial_t o_i.}$$

Spatial boundary terms vanish, so $Q'_i = Q_i$. The extra terms in continuity cancel because mixed derivatives commute. This is freedom in local charge representatives.

Which quantities stay fixed, and why?

In *every* homogeneous stationary GGE, $\langle \partial_x o_i \rangle = \langle \partial_t o_i \rangle = 0$. Thus both homogeneous density and current means are unchanged throughout the family: $F' = F$ and $A' = A$. On a periodic ring of length ℓ , $C_{ij} = \langle Q_i Q_j \rangle^c / \ell$; the same charges and state give the same C . Consequently $D' = A^2 C = D$.

The distinction between total weight and first moment matters. Set $R_{ij} = \langle o_i(x) q_j(0) \rangle^c$, $U_{ij} = \langle q_i(x) o_j(0) \rangle^c$, and $V_{ij} = \langle o_i(x) o_j(0) \rangle^c$ at equal time. Translation gives

$$S' = S + \partial_x R - \partial_x U - \partial_x^2 V, \quad C' = C, \quad E' - E = - \int dx R + \int dx U.$$

The first-moment correction need not vanish: integration by parts leaves integrals, not just boundaries. Appropriate decay is required in all these steps.

Why $\mathfrak{D}$ can change

If $\bar{o}_i = O_i(\bar{q})$ at leading order and $B_i^j = \partial O_i / \partial \bar{q}_j$, then

$$\bar{q}' = \bar{q} + B \partial_x \bar{q} + O(\partial_x^2), \quad -\partial_t \bar{o} = B A \partial_x \bar{q} + O(\partial_x^2).$$

The second relation uses Euler evolution $\partial_t \bar{q} = -A \partial_x \bar{q} + O(\partial_x^2)$. Diffusive evolution would add a second-gradient current term, beyond the coefficient being tracked. Reexpressing $F(\bar{q})$ in $\bar{q}'$ also contributes at first-gradient order. These changes can reshuffle $\mathfrak{D}$.

$\mathfrak{L}$ remains invariant under the hydrodynamic-projection assumption of Appendix C.1. Total time derivatives in the double current integral become endpoint terms; projection establishes a combined $O(1)$ correction, which changes neither its t^2 nor its t coefficient. Endpoint terms are not automatically bounded.

How PT supplies a compatible representative

The strong PT hypothesis is an antiunitary involution preserving H , momentum, every Q_i , and locality. Define $\tilde{q}_i(x, t) = Tq_i(-x, -t)T^{-1}$. Equation (2.29) implies $\tilde{q}_i - q_i = \partial_x b_i$ for a local b_i . Average the representatives:

$$q_i^{\text{PT}} = \frac{1}{2}(q_i + \tilde{q}_i) = q_i + \partial_x(b_i/2), \quad Tq_i^{\text{PT}}(x, t)T^{-1} = q_i^{\text{PT}}(-x, -t).$$

This preserves the charge and supplies (2.30). It is an existence construction; the full proper-gauge conditions are treated in Appendix C.

For the accompanying current define $\tilde{j}_i = Tj_i(-x, -t)T^{-1}$. Both reflected derivatives change sign in continuity. Comparing the transformed and original equations gives

$$\partial_x(\tilde{j}_i - j_i) = 0.$$

Thus $j_i^{\text{PT}} = (j_i + \tilde{j}_i)/2$ has the same divergence and obeys (2.31). Under the Appendix C locality property, the difference is an identity constant, irrelevant to connected correlations.

Quantum picture and the essential antiunitarity caution

We use the **Heisenberg picture**: ρ is fixed and stationary, while $q_i(x, t) = e^{iHt/\hbar}q_i(x, 0)e^{-iHt/\hbar}$. Stationarity does not make local operators time independent. A change of quantum picture and a PT symmetry transformation are separate operations.

PT transforms both states and observables: $\rho' = T\rho T^{-1}$, $O' = TO T^{-1}$. The observable conjugation follows from $O'T|\psi\rangle = TO|\psi\rangle$. For real GGE sources and PT-invariant charges, $\rho' = \rho$. Writing $T = UK$ with K complex conjugation gives

$$\langle TOT^{-1} \rangle_\rho = \text{Tr}(\rho U O^* U^\dagger) = \text{Tr}(\rho^* O^*) = \langle O \rangle_\rho^*.$$

Here the matrix star is entrywise conjugation. The state transformation is used through its invariance. Antiunitary conjugation *preserves* product order:

$$T(AB)T^{-1} = (TAT^{-1})(TBT^{-1}), \quad (AB)^\dagger = B^\dagger A^\dagger.$$

Therefore for the ordered correlation (2.11),

$$\boxed{S_{ij}(x, t)^* = S_{ij}(-x, -t).}$$

Hermitian densities need not have a real product correlation:

$$\langle AB \rangle^c = \frac{1}{2}\langle \{A, B\} \rangle^c + \frac{1}{2}\langle [A, B] \rangle.$$

Its imaginary part comes from the commutator. PT makes the real part reflection-even and the imaginary part reflection-odd. The rigorous direct moment consequences are

$$\boxed{E^* = -E, \quad N(t)^* = N(-t).}$$

Correction to our earlier shortcut: changing variables gives $E = -\int dx x S(-x, 0) = -E^*$, not directly $E = -E$. An additional reality condition is needed to deduce $E = 0$. It is automatic classically; real symmetrized correlations provide another convention, but the paper defines an ordered correlation. We have not established the required reality conditions for every such quantum moment.

The paper's PT-gauge conclusion and its payoff

The paper asserts

$$E = 0 \quad (2.33), \quad N(t) = N(-t) \quad (2.32).$$

The simple reflection proof of these statements needs real relevant moments; keep the qualification above. Accepting these PT-gauge results, match the two hydrodynamic branches:

$$\mathfrak{D}C = C\mathfrak{D}^\top \quad (2.34), \quad \boxed{D = A^2C, \quad \mathfrak{L} = \mathfrak{D}C.} \quad (2.35)$$

This is susceptibility-weighted symmetry; $\mathfrak{D}$ itself need not be symmetric. For positive definite C , the matrix $C^{-1/2}\mathfrak{D}C^{1/2}$ is symmetric in fluctuation-normalized coordinates.

Why the factor C ? $\mathfrak{D}$ describes broadening of density perturbations; C supplies the amount of equilibrium fluctuation being spread. In a scalar example,

$$\partial_t S + v\partial_x S = \frac{\mathfrak{D}}{2}\partial_x^2 S, \quad N(t) = Cv^2t^2 + C\mathfrak{D}t$$

for a centred initial packet of weight C . Its variance grows as $\mathfrak{D}t$, whereas the unnormalized moment grows as $C\mathfrak{D}t$. Thus $\mathfrak{L} = C\mathfrak{D}$. The usual coefficient ν in $\nu\partial_x^2 S$ is $\nu = \mathfrak{D}/2$.

6. A conserved current has no residual Onsager part [§2.5]

The special operator identity is

$$j_0(x, t) = q_1(x, t) \quad (2.36) \quad \implies \quad J_0 = \int dx j_0 = Q_1 \text{ conserved.}$$

Examples: mass current equals momentum density in Galilean systems; energy current equals momentum density in relativistic systems; XXZ energy current is conserved. Particle-number current requires the appropriate mass factor.

Preserving operator order,

$$K_{0j}(s) = \int dx \langle q_1(x, s)j_j(0, 0) \rangle^c = \langle Q_1 j_j(0, 0) \rangle^c = k_j,$$

independent of s . Equations (2.15)–(2.16) then give

$$\boxed{D_{0j} = k_j, \quad \mathfrak{L}_{0j} = 0.}$$

It is the Onsager coefficient that vanishes, not necessarily the Drude weight. Equation (2.35) yields $\mathfrak{D}_0^k C_{kj} = 0$, hence, on an invertible sector,

$$\boxed{\mathfrak{D}_0^i = 0 \quad \text{for every } i.} \quad (2.37)$$

The current has no first-gradient correction. But $\partial_t \bar{q}_0 + \partial_x \bar{q}_1 = 0$ still evolves q_0 : coupled q_1 dynamics can affect its profile. A zero constitutive row does not imply every coupled mode is undamped. Nor does a zero row of $\mathfrak{D}$ imply a zero column, since $\mathfrak{D} = \mathfrak{L}C^{-1}$ mixes columns.

7. What to remember before Section 3

1. **Goal:** calculate diffusion beyond Euler propagation in an interacting integrable system.
2. **Section 2's output:** current correlations determine $\mathfrak{L}$; static density fluctuations determine C ; in the paper's PT gauge, $\mathfrak{D} = \mathfrak{L}C^{-1}$.

3. **What remains:** compute those microscopic equilibrium correlations using quasiparticles. This is why §3 introduces thermodynamics and particle–hole form factors.
4. **Essential assumptions:** local relaxation and closure; a controlled long-wavelength expansion; decay sufficient for boundary terms; finite transport limits; the response input of Appendix B; hydrodynamic projection for gauge invariance; strong PT and the quantum-moment qualification above.
5. **Supporting branches:** response theory explains why mean closure controls correlations; moments identify transport coefficients; gauge fixing resolves local-density ambiguity. Each supports the same bridge.

Short review questions

1. What does conservation determine, and what does a current closure add?
2. Why is A a derivative across homogeneous GGEs, not a spatial gradient in one state?
3. Why must the persistent part D be subtracted before integrating to obtain $\mathfrak{L}$?
4. Why does negative-time evolution act on the second index?
5. Why can a gauge change preserve C but change E and $\mathfrak{D}$?
6. What additional condition turns $E = -E^*$ into $E = 0$?
7. Why does the bridge contain C , and on which side must its inverse appear?
8. Why can $j_0 = q_1$ give zero diffusion row while leaving nonzero transport?

Source cautions. The apparent E -factor discrepancy in (2.26)–(2.27), and the apparent q_0 versus q_1 index issue in (2.38), are recorded in the companion source-audit review. Neither is an author-confirmed erratum. The PT reality qualification is explained above so that it is not lost during review.